

Data Engineering Workflow Platform

Open Source Data Engineering Platform สำหรับสร้างและจัดการ Data Pipeline ตั้งแต่ Data Source → Data Lake → Transformation → Data Quality → Data Warehouse → Analytics → AI



1. Project Overview

Data Engineering Workflow Platform (DEWP) เป็นแพลตฟอร์มสำหรับสร้าง Data Engineering Workflow แบบครบวงจร โดยเน้นให้สามารถ

- Ingest ข้อมูล
- Transform ข้อมูล
- Validate ข้อมูล
- Orchestrate Pipeline
- Store ข้อมูล
- Monitor Pipeline
- Track Data Lineage
- ส่งข้อมูลต่อให้ BI, ML และ AI

แนวคิดหลัก:

```
DATA
↓
PIPELINE
↓
QUALITY
↓
WAREHOUSE
↓
ANALYTICS
↓
AI
```

2. Quick Start

2.1 Requirements

ติดตั้งเครื่องมือพื้นฐาน:

```
brew install git python uv  
brew install --cask docker
```

ตรวจสอบ:

```
git --version  
python3 --version  
uv --version  
docker --version  
docker compose version
```

3. Create Project

สร้าง Project:

```
mkdir data-engineering-workflow  
cd data-engineering-workflow
```

สร้าง Python Environment:

```
uv venv --python 3.13  
source .venv/bin/activate
```

ติดตั้ง Python Packages:

```
uv pip install \  
    pandas \  
    polars \  
    sqlalchemy \  
    psycpg2-binary \  
    requests \  
    python-dotenv \  
    pydantic \  
    dbt-postgres
```

4. Project Structure

สร้างโครงสร้างเริ่มต้น:

```
mkdir -p \  
dags \  
ingestion \  
transformation \  
quality \  
warehouse \  
dbt/models/staging \  
dbt/models/intermediate \  
dbt/models/marts \  
data/bronze \  
data/silver \  
data/gold \  
tests \  
scripts \  
notebooks \  
docs \  
monitoring
```

โครงสร้าง:

```
data-engineering-workflow/  
|  
├── dags/  
├── ingestion/  
├── transformation/  
├── quality/  
├── warehouse/  
├── dbt/  
│   ├── models/  
│   │   ├── staging/  
│   │   ├── intermediate/  
│   │   └── marts/  
├── data/  
│   ├── bronze/  
│   ├── silver/  
│   └── gold/  
├── tests/  
├── scripts/  
├── notebooks/  
├── docs/  
├── monitoring/  
├── docker-compose.yml  
└── requirements.txt
```

```
|— .env
|— .gitignore
|— README.md
```

5. Start Infrastructure

Development Infrastructure ใช้:

```
Docker
|
|— PostgreSQL
|— MinIO
|— Airflow
```

เริ่มระบบ:

```
docker compose up -d
```

ตรวจสอบ:

```
docker compose ps
```

หยุดระบบ:

```
docker compose down
```

6. First Pipeline

Pipeline ตัวแรกใช้ Sales Dataset:

```
CSV
↓
Extract
↓
Validate
↓
Bronze
```

```
↓
Transform
↓
Silver
↓
Data Quality
↓
Gold
↓
PostgreSQL
```

7. Create Sample Data

สร้างไฟล์:

```
data/bronze/sales.csv
```

ข้อมูลตัวอย่าง:

```
transaction_id,transaction_date,customer_id,product_id,quantity,unit_price
TX001,2026-09-01,C001,P001,2,100.00
TX002,2026-09-01,C002,P002,1,250.00
TX003,2026-09-02,C001,P003,3,150.00
TX004,2026-09-02,C003,P001,5,100.00
TX005,2026-09-03,C004,P002,2,250.00
TX006,2026-09-03,C002,P003,1,150.00
```

8. Run Transformation

ตัวอย่าง Python:

```
python transformation/sales.py
```

ผลลัพธ์:

```
data/
|
├─ bronze/
```

```
└─ sales.csv
└─ silver/
   └─ sales_clean.csv
└─ gold/
   └─ sales_daily.csv
```

9. Verify the Result

ตรวจสอบข้อมูล:

```
cat data/silver/sales_clean.csv
```

ตรวจสอบ Gold:

```
cat data/gold/sales_daily.csv
```

ตัวอย่างผลลัพธ์:

```
transaction_date,transaction_count,units_sold,revenue
2026-09-01,2,3,450
2026-09-02,2,8,750
2026-09-03,2,3,650
```

10. Verify PostgreSQL

เข้า PostgreSQL:

```
docker exec -it de-postgres \
psql -U dataeng -d datawarehouse
```

ตรวจสอบ Schema:

```
\dn
```

ตรวจสอบ Tables:

```
\dt raw.*
\dt staging.*
\dt mart.*
```

Query:

```
SELECT *
FROM mart.sales_daily;
```

11. Architecture

ระบบประกอบด้วย:

```
graph TD
    DS[Data Sources] --> I[Ingestion]
    I --> DL[Data Lake]
    DL --> B[Bronze]
    B --> T[Transformation]
    T --> S[Silver]
    S --> DQ[Data Quality]
    DQ --> G[Gold]
    G --> DW[Data Warehouse]
    DW --> BA[BI / Analytics / ML / AI]
```

flowchart TB

```
SOURCES["DATA SOURCES"]
INGEST["INGESTION"]
LAKE["DATA LAKE MinIO"]
BRONZE["BRONZE Raw"]
TRANSFORM["TRANSFORMATION Python / SQL / dbt / Spark"]
SILVER["SILVER Cleaned"]
```

```
QUALITY["DATA QUALITY"]
GOLD["GOLD Business Ready"]
WAREHOUSE["DATA WAREHOUSE PostgreSQL"]
CONSUMPTION["BI / Analytics / ML / AI"]

SOURCES --> INGEST
INGEST --> LAKE
LAKE --> BRONZE
BRONZE --> TRANSFORM
TRANSFORM --> SILVER
SILVER --> QUALITY
QUALITY --> GOLD
GOLD --> WAREHOUSE
WAREHOUSE --> CONSUMPTION
```

12. Data Workflow

flowchart TD

```
START(["START"])
EXTRACT["Extract"]
VALIDATE["Validate"]
BRONZE["Load Bronze"]
CLEAN["Clean"]
TRANSFORM["Transform"]
SILVER["Load Silver"]
QUALITY["Data Quality"]
GATE{"PASS?"}
QUARANTINE["Quarantine"]
GOLD["Build Gold"]
WAREHOUSE["Load Warehouse"]
PUBLISH["Publish"]
END(["SUCCESS"])
```

```
START --> EXTRACT
EXTRACT --> VALIDATE
VALIDATE --> BRONZE
BRONZE --> CLEAN
CLEAN --> TRANSFORM
TRANSFORM --> SILVER
SILVER --> QUALITY
QUALITY --> GATE
```

```
GATE -->|YES| GOLD
GATE -->|NO| QUARANTINE
```

```
GOLD --> WAREHOUSE
```

```
WAREHOUSE --> PUBLISH
PUBLISH --> END
```

13. Bronze / Silver / Gold

Project ใช้ **Medallion Architecture**

flowchart LR

```
SOURCE["SOURCE"]
BRONZE["BRONZE Raw Data"]
SILVER["SILVER Cleaned Validated"]
GOLD["GOLD Business Ready"]
```

```
SOURCE --> BRONZE
BRONZE --> SILVER
SILVER --> GOLD
```

Bronze

ข้อมูลต้นฉบับ:

```
Raw
Original
Immutable
Traceable
```

Silver

ข้อมูลผ่านการ:

```
Cleaning
Standardization
Deduplication
Validation
```

Gold

ข้อมูลพร้อมสำหรับ:

Analytics
Reporting
BI
ML
AI

14. Data Quality

ตัวอย่าง Quality Rules:

```
transaction_id
  → UNIQUE

customer_id
  → NOT NULL

quantity
  → > 0

unit_price
  → >= 0

transaction_date
  → VALID DATE
```

Architecture:

flowchart TD

```
DATA["Dataset"]
SCHEMA["Schema"]
NULL["NULL"]
DUP["Duplicate"]
TYPE["Data Type"]
RANGE["Range"]
FRESH["Freshness"]
BUSINESS["Business Rule"]
GATE{"Quality Gate"}
PASS["PASS"]
FAIL["FAIL"]
QUARANTINE["Quarantine"]
```

```
DATA --> SCHEMA
SCHEMA --> NULL
NULL --> DUP
```

```
DUP --> TYPE
TYPE --> RANGE
RANGE --> FRESH
FRESH --> BUSINESS
BUSINESS --> GATE

GATE --> |PASS| PASS
GATE --> |FAIL| FAIL

FAIL --> QUARANTINE
```

15. Apache Airflow

Airflow ทำหน้าที่เป็น Workflow Orchestrator

flowchart TD

```
DAG["Airflow DAG"]
INGEST["Ingestion"]
BRONZE["Bronze"]
SILVER["Silver"]
DBT["dbt"]
QUALITY["Quality"]
GOLD["Gold"]
WAREHOUSE["Warehouse"]
MONITOR["Monitoring"]

DAG --> INGEST
INGEST --> BRONZE
BRONZE --> SILVER
SILVER --> DBT
DBT --> QUALITY
QUALITY --> GOLD
GOLD --> WAREHOUSE
WAREHOUSE --> MONITOR
```

Airflow รับผิดชอบ:

- Scheduling
- Task Dependency
- Retry
- Failure Handling
- Monitoring
- Alerting

16. dbt

dbt ใช้สำหรับ SQL Transformation

```
Raw
↓
Staging
↓
Intermediate
↓
Mart
```

โครงสร้าง:

```
dbt/
├── models/
│   ├── staging/
│   │   └── stg_sales.sql
│   ├── intermediate/
│   │   └── int_sales.sql
│   └── marts/
│       └── fct_sales.sql
├── tests/
└── dbt_project.yml
```

คำสั่ง:

```
dbt debug
```

```
dbt run
```

```
dbt test
```

17. Data Lake

ใช้ MinIO เป็น Object Storage

```
MinIO
├── bronze
├── silver
└── gold
```

Data Flow:

flowchart LR

```
SOURCE["Data Source"]
BRONZE["MinIO Bronze"]
SILVER["MinIO Silver"]
GOLD["MinIO Gold"]
```

```
SOURCE --> BRONZE
BRONZE --> SILVER
SILVER --> GOLD
```

18. Data Warehouse

PostgreSQL:

```
PostgreSQL
├── raw
├── staging
├── intermediate
└── mart
```

ตัวอย่าง:

```
CREATE SCHEMA IF NOT EXISTS raw;
```

```
CREATE SCHEMA IF NOT EXISTS staging;
```

```
CREATE SCHEMA IF NOT EXISTS mart;
```

19. Data Lineage

flowchart LR

```
SOURCE["Source"]
BRONZE["Bronze"]
SILVER["Silver"]
GOLD["Gold"]
WAREHOUSE["Warehouse"]
DASHBOARD["Dashboard"]
AI["AI / RAG"]
```

```
SOURCE --> BRONZE
BRONZE --> SILVER
SILVER --> GOLD
GOLD --> WAREHOUSE
WAREHOUSE --> DASHBOARD
WAREHOUSE --> AI
```

สามารถติดตาม:

```
Source
  ↓
Transformation
  ↓
Dataset
  ↓
Warehouse
  ↓
Dashboard / AI
```

20. Monitoring

Monitoring ควรเก็บ:

```
Pipeline Status
Task Duration
Records Processed
Records Failed
Data Freshness
Quality Score
Error Rate
Retry Count
```

flowchart LR

```
PIPELINE["Pipeline"]
LOGS["Logs"]
METRICS["Metrics"]
ALERT["Alerts"]
DASHBOARD["Monitoring Dashboard"]
```

```
PIPELINE --> LOGS
PIPELINE --> METRICS
```

```
LOGS --> DASHBOARD
METRICS --> DASHBOARD
```

```
METRICS --> ALERT
```

21. Technology Stack

Layer	Technology
Language	Python
Query	SQL
Container	Docker
Orchestration	Apache Airflow
Transformation	dbt
Data Lake	MinIO
Warehouse	PostgreSQL
Processing	Pandas / Polars
Distributed Processing	Apache Spark
Quality	dbt Tests / Great Expectations
Lineage	OpenLineage
Catalog	DataHub
Monitoring	Prometheus / Grafana
Version Control	Git
CI/CD	GitHub Actions

22. Project Structure

```
data-engineering-workflow/  
├── dags/  
├── ingestion/  
├── transformation/  
├── quality/  
├── warehouse/  
  
├── dbt/  
│   ├── models/  
│   │   ├── staging/  
│   │   ├── intermediate/  
│   │   └── marts/  
│   ├── tests/  
│   └── dbt_project.yml  
  
├── data/  
│   ├── bronze/  
│   ├── silver/  
│   └── gold/  
  
├── monitoring/  
├── tests/  
├── notebooks/  
├── scripts/  
├── docs/  
  
├── docker/  
├── .github/  
│   └── workflows/  
  
├── docker-compose.yml  
├── requirements.txt  
├── .env.example  
├── .gitignore  
├── README.md  
└── LICENSE
```

23. Data Model

ตัวอย่าง Sales Data Warehouse:

erDiagram

```

CUSTOMER {
    string customer_id PK
    string customer_name
    string segment
}

PRODUCT {
    string product_id PK
    string product_name
    string category
    decimal price
}

DATE {
    int date_key PK
    date date
    int year
    int month
    int day
}

SALES {
    string transaction_id PK
    string customer_id FK
    string product_id FK
    int date_key FK
    int quantity
    decimal unit_price
    decimal total_amount
}

CUSTOMER ||--o{ SALES : buys
PRODUCT ||--o{ SALES : contains
DATE ||--o{ SALES : occurs

```

24. AI / RAG Integration

Data Engineering Platform สามารถเป็นฐานข้อมูลสำหรับ AI:

flowchart TB

```

DATA["Enterprise Data"]
PLATFORM["Data Engineering Platform"]
LAKE["Data Lake"]
WAREHOUSE["Data Warehouse"]
KNOWLEDGE["Knowledge Layer"]
RAG["RAG"]
AGENT["AI Agent"]

```

DATA --> PLATFORM

PLATFORM --> LAKE

PLATFORM --> WAREHOUSE

LAKE --> KNOWLEDGE

WAREHOUSE --> KNOWLEDGE

KNOWLEDGE --> RAG

RAG --> AGENT

รองรับ:

- Knowledge Base
- RAG
- Embedding Pipeline
- Vector Search
- AI Agent
- Document Processing
- Retrieval Evaluation

25. Distributed Processing

เมื่อข้อมูลมีขนาดใหญ่ สามารถเพิ่ม Apache Spark:

flowchart LR

SOURCE["Large Data"]

LAKE["Data Lake"]

SPARK["Apache Spark"]

SILVER["Silver"]

GOLD["Gold"]

WAREHOUSE["Warehouse"]

SOURCE --> LAKE

LAKE --> SPARK

SPARK --> SILVER

SILVER --> GOLD

GOLD --> WAREHOUSE

แนวทางเลือก Processing Engine:

Small Data

↓

Python / Pandas

Medium Data



Polars / SQL

Large Data



Apache Spark

26. Development Roadmap

Phase 01

Foundation



Phase 02

Data Ingestion



Phase 03

Data Lake



Phase 04

ETL / ELT



Phase 05

Airflow



Phase 06

dbt



Phase 07

Data Quality



Phase 08

Data Warehouse



Phase 09

Data Lineage



Phase 10

Data Catalog



Phase 11

Monitoring



Phase 12

CI/CD



Phase 13
Spark
↓
Phase 14
Production
↓
Phase 15
AI / RAG / Agent

27. Production Architecture

flowchart TB

USERS["Users / Applications"]

subgraph SOURCES["DATA SOURCES"]

DB["Databases"]

API["APIs"]

FILE["Files"]

EVENT["Events"]

end

subgraph PLATFORM["DATA ENGINEERING PLATFORM"]

INGEST["Ingestion"]

LAKE["Data Lake"]

BRONZE["Bronze"]

SILVER["Silver"]

QUALITY["Data Quality"]

GOLD["Gold"]

WH["Data Warehouse"]

LINEAGE["Lineage"]

CATALOG["Catalog"]

MONITOR["Monitoring"]

end

AIRFLOW["Apache Airflow"]

subgraph CONSUMPTION["CONSUMPTION"]

BI["BI"]

ANALYTICS["Analytics"]

ML["ML"]

AI["AI / RAG / Agent"]

end

DB --> INGEST

API --> INGEST

FILE --> INGEST

EVENT --> INGEST

INGEST --> LAKE
LAKE --> BRONZE
BRONZE --> SILVER
SILVER --> QUALITY
QUALITY --> GOLD
GOLD --> WH

WH --> BI
WH --> ANALYTICS
WH --> ML
WH --> AI

AIRFLOW -. -> INGEST
AIRFLOW -. -> BRONZE
AIRFLOW -. -> SILVER
AIRFLOW -. -> GOLD
AIRFLOW -. -> QUALITY
AIRFLOW -. -> WH

QUALITY --> LINEAGE
WH --> LINEAGE
LINEAGE --> CATALOG

AIRFLOW -. -> MONITOR
QUALITY -. -> MONITOR
WH -. -> MONITOR

USERS --> CONSUMPTION

28. Production Checklist

Infrastructure

- ☐ Docker
- ☐ PostgreSQL
- ☐ MinIO
- ☐ Airflow

Pipeline

- ☐ Ingestion
- ☐ Bronze
- ☐ Silver
- ☐ Gold
- ☐ Warehouse

Quality

- ☐ Schema Validation
- ☐ NULL Check
- ☐ Duplicate Check
- ☐ Business Rules
- ☐ Freshness

Operations

- ☐ Scheduling
- ☐ Retry
- ☐ Logging
- ☐ Monitoring
- ☐ Alerting

Governance

- ☐ Data Lineage
- ☐ Data Catalog
- ☐ Metadata
- ☐ Data Ownership

Engineering

- ☐ Git
- ☐ Tests
- ☐ CI/CD
- ☐ Docker
- ☐ Documentation

Advanced

- ☐ Spark
- ☐ Distributed Processing
- ☐ ML Pipeline
- ☐ RAG
- ☐ AI Agent

29. Final Vision

flowchart TB

```
DATA["DATA"]
INGEST["INGEST"]
STORE["STORE"]
PROCESS["PROCESS"]
```

```
VALIDATE["VALIDATE"]  
ORCHESTRATE["ORCHESTRATE"]  
GOVERN["GOVERN"]  
MONITOR["MONITOR"]  
ANALYTICS["ANALYTICS"]  
AI["AI"]
```

```
DATA --> INGEST  
INGEST --> STORE  
STORE --> PROCESS  
PROCESS --> VALIDATE  
VALIDATE --> ORCHESTRATE  
ORCHESTRATE --> GOVERN  
GOVERN --> MONITOR
```

```
MONITOR --> ANALYTICS  
ANALYTICS --> AI
```

Vision:

Build → Automate → Validate → Observe → Govern → Scale → AI

Data Engineering Workflow Platform ถูกออกแบบให้เริ่มจาก Pipeline ขนาดเล็กบนเครื่อง Local ได้ทันที และสามารถขยายไปสู่ Data Platform, Distributed Processing และ AI Data Infrastructure ได้ในอนาคต.

30. License

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